

Systematic analysis of life cycle assessment tools in supply chain management

Busola Akintayo^{a,b,c,*}, Angel M. Gento^b, Damilola Caleb Akintayo^d, Yesica Pino^b,
Oludolapo Olanrewaju^{a,c}

^a Department of Industrial Engineering, Durban University of Technology, Durban 4001, South Africa

^b Escuela de Ingenierías Industriales, Universidad de Valladolid, Paseo Prado de la Magdalena 3-5, 47011 Valladolid, Spain

^c Institute of Systems Science, Durban University of Technology, Durban 4001, South Africa

^d School of Chemistry & Physics, Faculty of Agriculture, Engineering & Science, Westville Campus, University of KwaZulu Natal, Durban 4010, South Africa

ARTICLE INFO

Keywords:

LCA
Impact assessment
SCM
Industry
PRISMA
SimaPro
Ecoinvent
ReCiPe

ABSTRACT

The operational integration of Life Cycle Assessment (LCA) within Sustainable Supply Chain Management (SCM) is critical for driving global decarbonisation, yet its academic literature remains methodologically fragmented. This study conducts a systematic literature review and bibliometric analysis of 96 peer-reviewed articles to map the current state, structural boundaries, and systemic gaps within the LCA-SCM discipline. The empirical findings reveal a highly uneven sectoral landscape, with agri-food systems emerging as the dominant frontier representing 31.2% of the surveyed corpus, followed by manufacturing and waste management, whilst carbon-intensive sectors like logistics remain underexplored. Methodologically, the field is governed by powerful regional path dependencies, with SimaPro and Ecoinvent serving as the primary technical platforms characterising over three-quarters of the analysed literature. Furthermore, this review exposes a widespread crisis of methodological rigour and transparency across global value chains: 55% of the examined studies truncate their system boundaries to a cradle-to-gate scope, functional units are frequently conflated with simple mass-based declared units, and 47% completely omit both sensitivity and uncertainty analyses. This lack of statistical validation severely undermines the auditable reliability of environmental disclosures. To transition LCA from a retrospective diagnostic tool into a proactive sourcing enabler, this study provides a cross-sectoral decision guide for practitioners and outlines a strategic future research agenda. This framework prioritises the standardisation of allocation metrics, the expansion of Social LCA, the institutionalisation of dynamic inventory modelling, and the deployment of Industry 4.0 digital technologies like blockchain and artificial intelligence.

1. Introduction

Sustainable development, as conceptualized by the Brundtland Report (Keeble, 1988), seeks to meet the needs of the present without compromising the ability of future generations to meet their own. Over the past decade, the urgency of achieving sustainability has intensified, driven by climate change, resource depletion, and economic disparities (Yamaguchi et al., 2023). In response to the urgent need to mitigate the consequences of continuous industrialisation, the United Nations' Sustainable Development Goals (SDGs) were introduced in 2015 (United Nations, 2015), comprising 17 goals and 169 targets aimed at integrating sustainability into economic, environmental, and social

structures. The SDGs emphasize key areas such as responsible production, climate action, clean energy, and sustainable industry, encouraging collaboration between governments, businesses, and civil society (Leal Filho et al., 2019).

As industries and economies adopt sustainability principles, Life Cycle Assessment (LCA) has emerged as a critical method for evaluating and improving environmental performance (Hauschild et al., 2018). Regulated by the ISO 14040 standards, LCA provides a structured, quantitative framework consisting of four main phases: goal and scope definition, life cycle inventory analysis (LCI), life cycle impact assessment (LCIA), and interpretation (International Organization for Standardization, 2006a; Arvanitoyannis, 2008). This widely

* Corresponding author at: Department of Industrial Engineering, Durban University of Technology, Durban 4001, South Africa.

E-mail addresses: olagunjubusola52@gmail.com (B. Akintayo), angel.gento@uva.es (A.M. Gento), damilolaakintayo@141gmail.com (D.C. Akintayo), yesica.pino@uva.es (Y. Pino), oludolapoo@dut.ac.za (O. Olanrewaju).

<https://doi.org/10.1016/j.clscn.2026.100371>

Received 9 September 2025; Received in revised form 22 June 2026; Accepted 1 September 2026

Available online 2 September 2026

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recognised methodology facilitates the evaluation of environmental impacts and potential consequences of products, processes, and organisations across their entire life cycle—from raw material extraction to production, distribution, use, and final disposal (International Organization for Standardization, 2006b; Curran, 2012). By enabling the identification of environmental hotspots, LCA informs robust strategies for mitigating adverse effects. In the realm of supply chain management, it serves as an essential tool for assessing and enhancing the sustainability of interconnected processes, and has been extensively utilised in various decision-making contexts, including policy formulation and corporate sustainability evaluations (Di Maria and Micale, 2014; Butera et al., 2015; Akintayo et al., 2024).

Supply Chain Management (SCM) plays a crucial role in advancing sustainability, particularly when examined from a life cycle perspective (Fet et al., 2023). Though not originally sustainability-focused, SCM shares structural traits with LCA, enabling simultaneous improvements in environmental and economic performance. According to Blass and Corbett (2018), SCM promotes the adoption of tools like LCA and Life Cycle Costing (LCC), thereby integrating sustainability into day-to-day business practices. Emerging from logistics in the late 1980s, SCM evolved to manage not just internal flows but also upstream and downstream relationships. Christopher (2016) defined logistics as the strategic coordination of procurement, movement, and storage to maximise profitability. SCM expands this scope to oversee the entire supply chain, aiming to deliver higher customer value at reduced costs through effective relationship and resource management. Crucially, modern supply chains account for up to 90% of a system's total carbon footprint (Zhang et al., 2023a, 2023b), making them the primary site at which environmental impacts are generated and where LCA-informed strategies are most urgently required.

The interorganisational nature of SCM presents unique challenges. Coyle et al. (1992) identify key areas of concern, including inventory management, cost minimisation for end customers, efficient information exchange, collaboration fostering, and the balancing of risks and rewards among supply chain partners. As the concept of circular supply chains gains traction, these challenges become increasingly complex. De Angelis et al. (2018) note that transitioning to circular supply chains fundamentally alters interaction patterns, material flows, and value exchanges, necessitating enhanced collaboration, innovative business models, and rigorous strategic planning.

A life cycle perspective inherently aligns with supply chain thinking, as the costs and environmental impacts associated with product development, production, use, and disposal correspond directly to different supply chain stages. Each actor, from component manufacturers to final users, contributes to both operational service levels and cumulative environmental footprints. Since the early 2000s, SCM's potential as a driver for sustainability has gained significant attention, leading to the development of Sustainable Supply Chain Management (SSCM) (Carter and Liane Easton, 2011; Saeed and Kersten, 2019). Carter and Rogers (2008) define SSCM as the strategic, transparent integration of social, environmental, and economic goals into key business processes to enhance long-term economic performance at both the company and supply chain levels.

Despite this clear potential, significant barriers continue to hinder the effective operationalisation of LCA within SCM. These include the limited standardisation of tools and datasets, sector-specific application constraints, methodological inconsistencies across empirical studies, and the near-absence of social LCA dimensions in supply chain assessments. Consequently, the utility of existing evidence for cross-sector decision-making remains constrained (Finnveden et al., 2009; Owsianiak et al., 2014). Furthermore, as organisations transition toward circular supply chains, LCA is increasingly expected to capture circular dynamics; however, most applications remain linear in scope. While existing methodologies are vital for energy and bio-based sectors where inputs and emissions evolve continuously, dynamic LCA models that track technological change and temporal variability remain

underdeveloped within the supply chain context.

1.1. Rationale for study

Several studies have explored the application of LCA in specific supply chain contexts, revealing diverse methodological practices and environmental insights. dos Muchangos et al. (2025) conducted a systematic review of LCA and Environmental Footprint (EF) methodologies in the global coffee value chain, analysing 87 studies and identifying five key research clusters, including by-products and waste management. They found that electricity and fuel consumption were major contributors to CO₂ emissions, whilst standardisation issues limited cross-study comparisons. Water footprint was the most frequently analysed EF, and coffee by-products were investigated for renewable and biomedical uses. Similarly, within the agri-food sector, Spina et al. (2024) focused on Circular Economy (CE) indicators in the olive oil industry, identifying metrics like the Material Circularity Indicator (MCI) and frameworks such as Environmental Product Declarations (EPD) and LCA, emphasising the need for multi-stakeholder collaboration to enhance circular practices. Complementing this, Rapa and Ciano (2022) reviewed LCA studies on olive oil production covering farming, extraction, packaging, and waste treatment. They observed that farming exerted the greatest environmental impact due to inputs like fertilisers and fuel, noting that carbon sequestration in olive groves is often underestimated in sustainability assessments, and advocated harmonised LCA methodologies to improve comparability. Shifting the focus to the livestock sector, Bhatt and Abbassi (2021) analysed LCA application in the sheep sector, considering stages from farm inputs to emissions. Methane emissions accounted for 50–75% of total greenhouse gases (GHGs), with substantial variability across farming systems, highlighting the acute need for consistent LCA approaches to enhance accuracy in environmental impact evaluations.

Overall, these systematic reviews reveal a consistent and consequential pattern of fragmentation when considered collectively, sharing three structural limitations. Firstly, each is confined to a single product or sector, making it difficult to draw cross-sectoral lessons regarding how LCA software, datasets, and tools are applied, or whether choices made in one sector are transferable to another. Secondly, none of the studies systematically maps methodological decisions (such as software selection, dataset usage, LCIA methods, system boundaries, and functional units) against industry contexts in a way that allows for systematic benchmarking. Thirdly, and most critically for SCM, none analyses how LCA findings are connected to core supply chain decisions such as procurement criteria, supplier selection, or environmental disclosure. The result is a body of literature that presents what LCA has found, but not how it is incorporated into the operational logic of supply chains. In this context, identifying and overcoming specific operational and methodological barriers is essential for successfully integrating life cycle assessment frameworks within sustainable industrial practices (Kaswan et al., 2020; Singh et al., 2022).

These limitations define the research gaps that the present study is designed to address: namely, the absence of a cross-sectoral synthesis of LCA tools within SCM, the lack of comparative analysis of methodological choices across industries, and the absence of a unified evidence base connecting these choices to operational implications. To address these gaps, this study aims to achieve three primary objectives. Firstly, it evaluates the prevalence and distribution of LCA tools—specifically software, datasets, and LCIA methods—applied within supply chain research across diverse industries. Secondly, it examines the specific LCIA approaches used in these studies, including system boundaries, functional units, and uncertainty analysis practices. Thirdly, the study identifies the specific industries in which LCA tools are most widely applied within supply chain contexts, ultimately assessing the implications of these observed patterns for sustainable supply chain decision-making.

1.2. Significance of study

This study fills a critical research gap by systematically analysing how LCA methodologies are applied across supply chains in diverse industries. While the existing literature focuses on sector- or product-specific cases, this work offers a holistic, cross-sectoral perspective on integrating LCA into SCM. It harmonises industry applications, system boundaries, and methodological trends, revealing opportunities for standardising and improving LCA tools, software, datasets, and LCIA methods. Through this, the study identifies best practices and inconsistencies, thereby advancing reproducibility, decision-making, and comparability in sustainable practices. Furthermore, it offers practical and policy-relevant insights, highlighting methodological gaps and guiding sustainable supply chain design. Serving as a robust reference point for businesses, policymakers, and practitioners, it supports environment-centred strategies in production, distribution, and procurement. Ultimately, the study contributes to the achievement of the United Nations' Sustainable Development Goals, particularly SDG 12 (Responsible Consumption and Production) and SDG 13 (Climate Action).

The remainder of this paper is structured as follows. Section 2 presents the methodology, including the PRISMA 2020 process, inclusion and exclusion criteria, and the data extraction approach. Section 3 provides an overview of the article pool and key bibliometric patterns on LCA in the context of SCM; Section 4 details the empirical results, whilst Section 5 develops the corresponding discussion. Section 6 identifies critical research gaps and future directions, including a structured decision guide for different stakeholder types. Finally, Section 7 summarises the conclusions, implications, and recommendations for both practice and policy, supplemented by the technical materials compiled in the Appendix.

2. Methodology

This study utilises a Systematic Literature Review (SLR) as its primary methodology, systematically supplemented by basic bibliometric mapping to contextualise the academic landscape of LCA-SCM research. Systematic reviews offer a recognised, rigorous approach for synthesising research evidence with transparency and reproducibility (Jesson et al., 2011). Following the PRISMA 2020 guidelines (Page et al., 2021), an SLR allows for a methodical and comprehensive analysis of scholarly studies to tackle defined research questions with transparency and rigour. To improve analytical depth, this study integrates quantitative and qualitative techniques. The overarching methodological framework is adapted from Visentin et al. (2020), which effectively combines systematic review protocols, bibliometric data visualization, and quantitative industry analysis to map the application of Life Cycle Assessment (LCA) in Supply Chain Management (SCM), uncover methodological patterns, and highlight research gaps.

2.1. Search strategy and selection criteria

The Scopus database was selected as the sole source for this review. Scopus was chosen because it represents one of the largest multidisciplinary academic databases, offering comprehensive coverage of peer-reviewed literature, along with robust citation, co-citation, and meta-data tools that effectively support systematic and bibliometric evaluations (Donthu et al., 2021). The systematic search was conducted using the following query string applied to the title, abstract, and keyword fields:

TITLE-ABS-KEY ("Life Cycle Assessment" OR "LCA") AND ("supply chain*" OR "supply chain management") AND ("tools" OR "software" OR "application") AND ("industry*" OR "sector").

The deliberate inclusion of the terms 'tools', 'software', and 'application' reflects a precise scoping decision. This review is specifically concerned with the operationalised use of LCA within supply chains;

that is, studies in which specific LCA software, datasets, or LCIA methods were applied to resolve a supply chain problem. Without these limiters, the search would return a substantially larger and more dispersed body of literature, including purely conceptual frameworks and policy discussions where LCA is referenced but not explicitly implemented. Restricting the scope to studies where LCA tools were clearly applied enables meaningful methodological benchmarking.

To ensure absolute transparency and maintain scholarly reliability, specific boundary-setting measures were employed during the screening process, as summarised in Table 1.

No restriction was placed on the publication year, thereby recognising the relatively recent emergence of LCA-SCM integration as a distinct field and ensuring the capture of foundational studies alongside current developments. The search covered all publications available in the database through March 2025.

2.2. Screening process and bibliographic data analysis

The screening workflow followed the four-stage PRISMA framework comprising Identification, Screening, Eligibility, and Inclusion. The step-by-step selection process is illustrated in the schematic flow chart in Fig. 1. During Stage 1 (Identification), the initial database search retrieved 500 records from Scopus. In Stage 2 (Screening), these records were evaluated by title and abstract against the established selection criteria, resulting in the exclusion of 302 records. Specifically, 151 records were excluded as non-journal document types (conference papers, book chapters, reports, or theses), five were non-English publications, and 146 did not apply LCA as a primary method or lacked an identifiable LCA tool, leaving 198 records for full-text assessment. During Stage 3 (Eligibility), a rigorous full-text review of these 198 articles led to the exclusion of a further 102 papers that did not meet the desired analytical scope, with reasons carefully recorded in line with PRISMA 2020 requirements. Consequently, in Stage 4 (Inclusion), a final sample of 96 peer-reviewed journal articles was selected to form the empirical basis for the analyses presented in Sections 4 and 5.

Following this data cleaning process, a two-stage analytical approach was implemented to explore the intellectual landscape of LCA within the context of SCM (Zupic and Čater, 2015; Donthu et al., 2021). The first phase involved examining publication trends over time to trace the trajectory of scholarly attention in LCA-SCM integration, identifying potential inflection points and displaying patterns of growth that indicate rising academic interest. The second stage involved a complementary bibliometric analysis performed using VOSviewer version 1.6.20 (Van Eck and Waltman, 2009).

VOSviewer creates graphical networks of bibliographic data, where the closeness between items (such as articles, journals, or keywords) reflects the strength of their co-occurrence or co-citation relationships

Table 1

Methodological framework: inclusion and exclusion criteria for study selection.

Criterion	Inclusion Criteria	Exclusion Criteria
Document Type	Peer-reviewed journal articles only	Conference papers, book chapters, reports, grey literature, and theses
Language	English-language publications only	Non-English articles
Database Source	Scopus-indexed journals	Articles not indexed in Scopus
Relevance & Scope	LCA applied as a primary methodology within a supply chain context	Articles mentioning LCA or supply chains only in passing; purely conceptual or policy papers with no operational LCA application
Tool Utilisation	Study explicitly employs or discusses a specific LCA software, dataset, or LCIA method	Studies utilising no identifiable LCA tool, software, or dataset

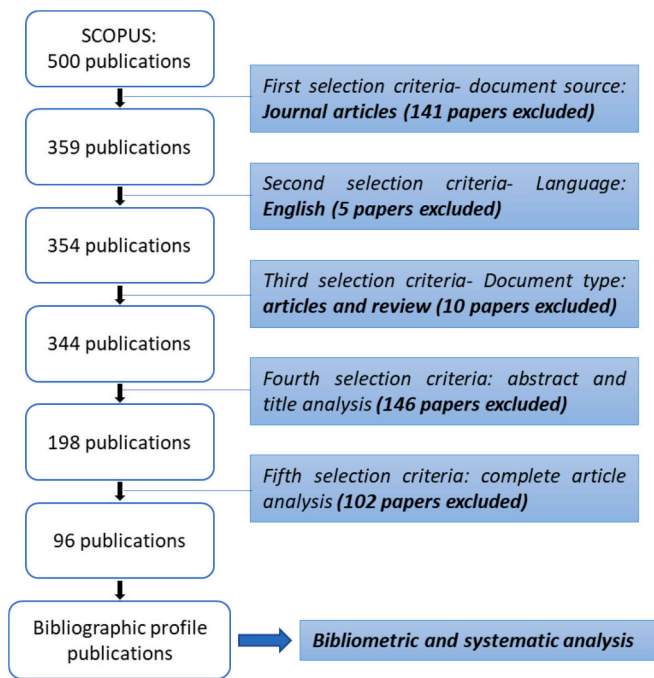


Fig. 1. Schematic flow chart of the literature selection process.

(Van Eck and Waltman, 2009). Items that are close to each other are considered more strongly linked, and clusters of related nodes emerge automatically, providing visual clarity into the structure of the research area. Each node differs in size based on parameters such as citation counts, average citations, or the number of publications, demonstrating the relative influence of that item. Colour-coded clustering further differentiates connected research fields, while density visualisations emphasize zones of concentrated scholarly activity, especially those journals and themes that dominate the citation landscape. Publication trends and journal co-citation outputs are presented in a condensed form within Section 3 as background context, whilst detailed network maps (co-authorship clusters, country collaboration networks, and keyword co-occurrences) are compiled in Supplementary Appendix B.

Finally, a quantitative industry distribution analysis was performed by coding and categorising the 96 articles based on their operational context. This step identifies the industries with the highest application of LCA tools within supply chain research, providing sector-specific knowledge on implementation patterns and facilitating the understanding of practical applications across sectors. Studies were assigned to one of nine primary categories: agri-food, manufacturing, energy and utilities, waste management, bioenergy and biofuels, construction, logistics and transportation, policy-oriented, or cross-sectoral. Frequency analysis was subsequently applied to determine the proportion of studies per category, with the primary application context guiding the classification in instances where studies spanned multiple sectors.

3. Overview of reviewed studies and key metrics

This section establishes the empirical and contextual foundation of the systematic review by mapping the macro-structural trends of the final 96 selected articles. Before dissecting the sector-specific technical configurations of Life Cycle Assessment (LCA) tools, it is vital to trace how this interdisciplinary domain has grown over time, where the core knowledge is published, and how geographic collaboration shapes research outputs. Consequently, the following subchapters provide a condensed analysis of the temporal evolution, leading academic journals, and global distribution patterns of the compiled literature, serving as a baseline context for the granular supply chain benchmarking that

follows.

3.1. Temporal distribution and publication evolution

The integration of Life Cycle Assessment (LCA) tools within Supply Chain Management (SCM) has evolved from a niche academic intersection into an indispensable domain of sustainability science. A chronological analysis of the 96 reviewed articles (with the search window closed in March 2025), as illustrated in Fig. 2, demonstrates a distinct multi-stage evolutionary trajectory. Between 2000 and 2010, research activity remained sparse and stagnant, typically averaging fewer than five documents per annum, characterised by foundational, static conceptualisations where LCA and logistics operated as isolated paradigms. A modest inflection point occurred around 2011, but it was from 2018 onwards that scholarly output experienced an exponential expansion, culminating in a remarkable publication peak in 2024 with over 40 documents, and maintaining substantial momentum up to early 2025.

This rapid acceleration in publication volume has created significant fragmentation in tool application, software choices, and data quality across the literature. Because supply chains account for up to 90% of a system's total environmental burden, this sudden influx of diverse research directly validates the critical necessity of the present systematic review. This section serves to bring methodological order, harmonisation, and benchmarking to a rapidly expanding but highly fragmented research landscape.

3.2. Leading journals and disciplinary fields

Identifying key journals that regularly publish studies on the combination of SCM and LCA is vital for researchers aiming to disseminate high-impact findings, and for funding bodies seeking reliable and high-visibility channels. Prior research indicates that citation metrics and journal reputation frequently serve as indicators of scholarly impact (Bornmann and Daniel, 2008; Tahamtan et al., 2016). Only five journals contributing to SCM-based LCA literature met the criteria of at least five documents and three citations per journal as detailed in Table 2.

These findings offer pragmatic guidance for authors, especially early-career researchers, in choosing journals that are most appropriate for disseminating their work to a wider and more engaged audience. It also highlights the value of targeting journals that are well-integrated into the core citation network of SCM-LCA scholarship. Whereas factors such as open-access availability, institutional prestige, and article novelty affect citation rates (Zhang et al., 2021; Mammola et al., 2022), publishing in highly interconnected journals can substantially increase scholarly visibility. Detailed bibliometric network maps, including peer co-authorship clusters, country-level collaboration networks, and keyword co-occurrence visualisations, are provided in Supplementary Appendix B.

3.3. Geographic distribution of research production

Scientific output and impact are increasingly shaped by international collaboration. Cross-border research enhances shared expertise, utilises advanced research infrastructure, and facilitates access to funding, resulting in a significant increase in the visibility and citation impact of scholarly publications (Glänzel and Schubert, 2004; Wagner et al., 2017). Within this domain, global research production is characterised by distinct collaborative networks where geographic, economic, and linguistic factors heavily shape partnership patterns (Katz and Martin, 1997; Leydesdorff and Wagner, 2008).

An analysis of publishing trends reveals that the United States is the top producing country, followed closely by Italy and the United Kingdom. While China exhibits slightly fewer documents within the reviewed corpus, its substantial citation impact and strong international connectivity indicate a high degree of global engagement. Conversely, other producing nations such as Portugal and Malaysia demonstrate

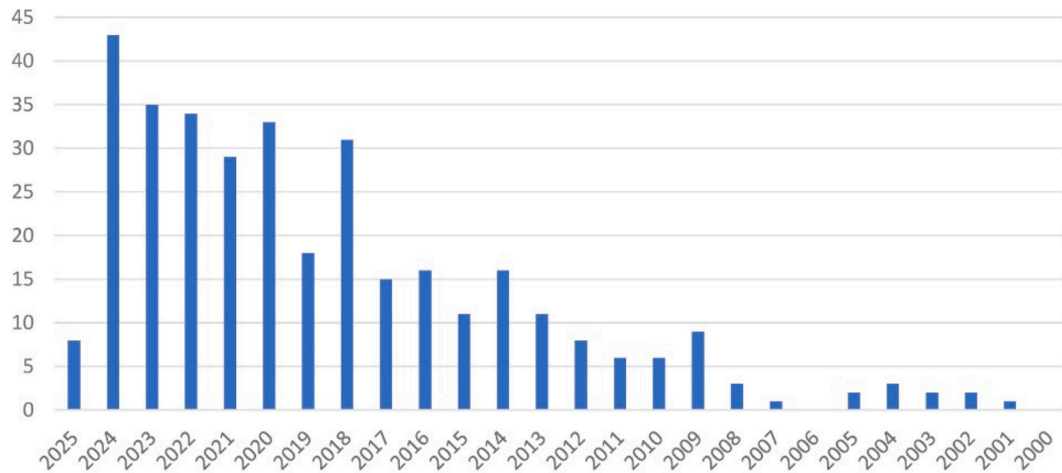


Fig. 2. Annual scientific output of SCM-based LCA.

Table 2

Bibliometric analysis of the top research journals.

Source	Documents	Citations	Total link strength ¹
Journal of Cleaner Production	18	685	121
International Journal of Life Cycle Assessment	10	516	72
Science of the Total Environment	8	284	44
Sustainability (Switzerland)	8	208	49
Resources, Conservation and Recycling	6	125	52

¹ Total link strength measures the cumulative strength of co-citation links between the respective journals within the bibliometric network.

lower collaborative interaction despite their active contributions.

Geopolitical and regional alignments significantly influence these structures. For instance, European research networks are highly integrated, with countries like Italy and the United Kingdom acting as central collaborative hubs. This strong intra-European and Anglo-European partnership model reflects the integration of researchers into high-impact international consortia, such as EU Horizon projects. Furthermore, even amidst broader geopolitical competition, strategic cross-border cooperation remains highly developed in high-impact scientific fields, as evidenced by the intense collaborative links between major global economies like the United States and China (Leydesdorff and Wagner, 2008).

The detailed bibliometric network maps, including the specific country-level co-authorship clusters, regional groupings, and the precise threshold criteria utilised for the technical visualization, are provided in Supplementary Appendix B.

4. Methodological practices in SCM-LCA

This section provides a comprehensive audit of the core methodological frameworks, tools, and structural parameters that govern the operationalisation of Life Cycle Assessment (LCA) within Supply Chain Management (SCM) literature. By synthesizing data from the final pool of 96 articles, the following subsections evaluate the technical architecture of current research, mapping software dependencies, dataset selection patterns, and the configuration of reference units and system boundaries. Furthermore, it addresses the statistical rigour of the field by scrutinising the prevalence of sensitivity and uncertainty analyses. This systemic benchmarking serves to expose critical trade-offs between methodological choices and their practical utility for sustainable supply chain decision-making.

4.1. Software and database selection: path dependencies and operational implications

The application of LCA tools across industrial sectors and the subsequent choice of software and data sources mirror the evolving priorities of sustainability science, revealing strong methodological and geographic alignments. As illustrated in Fig. 3, the reviewed literature exhibits a clear sectoral concentration, led predominantly by the agri-food industry, which accounts for 31.2% of the 96 reviewed studies. This is closely followed by materials and manufacturing, whilst sectors associated with the Circular Economy (CE)—such as waste management and bioenergy & biofuels—together account for 22.9% of the corpus, emphasising end-of-life processing, material recovery, and renewable resource utilisation (Cherubini et al., 2009; Tonini and Astrup, 2012). Conversely, critical carbon-intensive sectors such as logistics and transportation (8.3%) or energy and utilities (7.3%) remain relatively understudied within this framework (Spath and Mann, 2000; Koroneos et al., 2008).

These sector-specific requirements deeply influence and couple with the choice of technical tools. As detailed in Fig. 4, SimaPro stands out as the de facto standard in the field, being utilised in over 50% of the analysed literature due to its sophistication and compatibility with core databases, particularly in European and academic settings (Finnveden et al., 2009; Owsianiak et al., 2014; Speck et al., 2016; Fritter et al., 2020; Rybaczewska-Błażejowska and Jezierski, 2024). In contrast, openLCA and GaBi are used less frequently but remain prominent; GaBi is chosen for its process-based modelling strengths, whereas openLCA's attraction lies in its open-source nature, often deployed in developing countries or lower-resource research settings. It is critical to note that 14% of the studies fail to explicitly disclose the software version or tool used, representing a significant gap in methodological transparency.

Regarding inventory data, a profound dependency on a single data source is observed across these industries. As shown in Fig. 5, the Ecoinvent database dominates the choice of dataset with approximately 78% usage, either exclusively or in combination with primary data, due to its comprehensive unit-process data and broad software compatibility (Weidema et al., 2013). Conversely, datasets such as Agribalyse or GaBi make up a minimal portion of the literature. Furthermore, primary-only data represents less than 5% of the corpus, indicating that most LCA applications in SCM rely on secondary or hybrid approaches (about 20%) due to the inherent scale and complexity of multi-stage supply chains (Brandao et al., 2011).

Rather than being random, these combined preferences for software, datasets, and impact assessment frameworks demonstrate a powerful regional and institutional path dependency. For instance, all studies utilising the North American TRACI method align strictly with US EPA

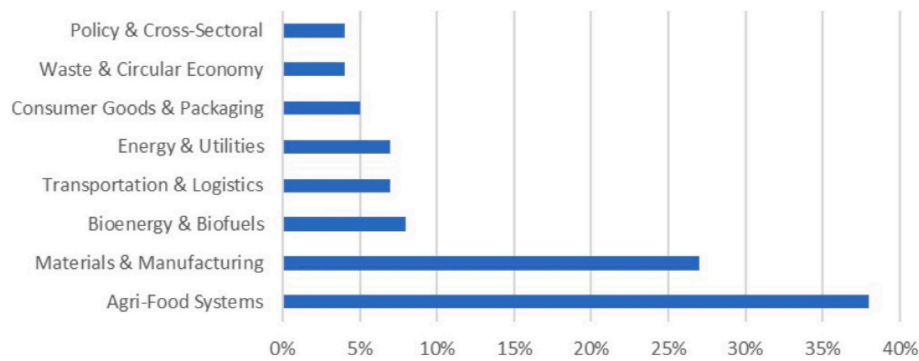


Fig. 3. Industries application of LCA within the 96 reviewed studies.

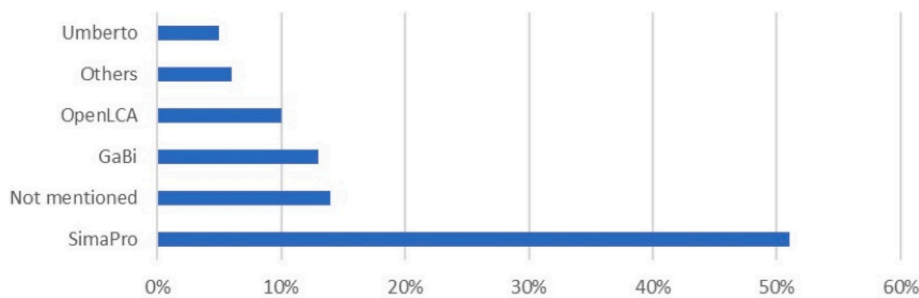


Fig. 4. LCA software tools utilised within the 96 reviewed studies.

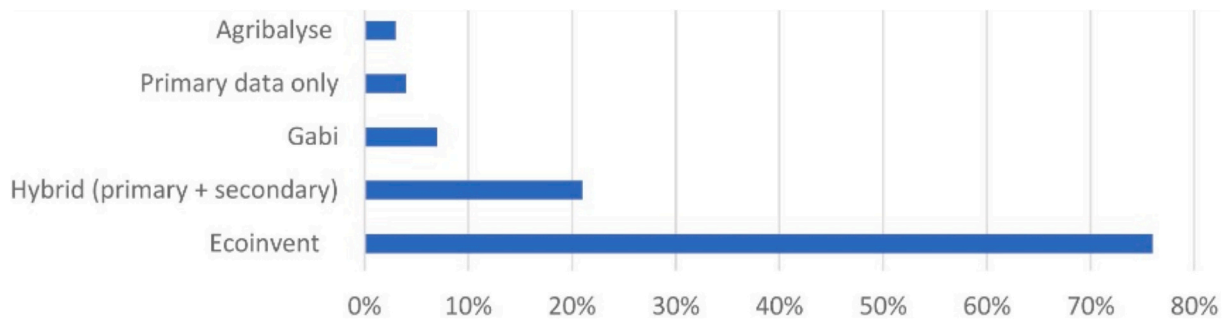


Fig. 5. LCA dataset utilised within the 96 reviewed studies.

guidelines, whereas European and Asian studies largely rely on the ReCiPe method (utilised in 45% of all studies, rising to 70% in agri-food and 75% in policy sectors) (Goedkoop et al., 2009; Del Borghi et al., 2014; Owsianiak et al., 2014; Huijbregts et al., 2017; Notarnicola et al., 2017). This regional lock-in is further evidenced by a strong software-tool coupling: a substantial 91% of ReCiPe-based analyses were conducted exclusively within SimaPro, which may limit methodological flexibility.

This institutional path dependency creates severe operational implications for supply chain research in emerging economies; while researchers in lower-resource settings frequently adopt openLCA due to its free availability, the acute lack of domestic or localized inventory databases compels them to rely on generic European data from Ecoinvent. Consequently, these assessments fail to reflect local energy mixes, domestic transport efficiencies, and regional production realities, thereby distorting the true environmental footprint of their supply chains (Bare, 2002; Bare, 2011; Bare et al., 2012; Rybaczewska-Błazejowska and Jezierski, 2024).

4.2. Functional units vs. declared units in supply chain contexts

4.2.1. Methodological foundations and quantitative distribution

The selection of a reference unit establishes the mathematical basis for inventory aggregation, environmental allocation, and comparative assertions across multi-tiered supply chain networks. Analysis of the 96 reviewed articles reveals a substantial preference for mass-based units, which are employed in over 50% of the surveyed literature. Volume- and energy-based units collectively account for 21% of the studies, typically constrained to liquid product systems and energy-intensive manufacturing processes, respectively. The remaining 18% comprises custom, sector-specific metrics, such as service hours or combined multi-functional units. The distribution of these metrics across key industrial supply chains is detailed in Table 3.

4.2.2. The functional unit vs. declared unit dichotomy

A critical methodological distinction exists between Functional Units (FUs) and Declared Units (DUs) under the ISO 14040/14044 standards. This distinction is frequently blurred in supply chain literature, introducing structural ambiguity into cross-study comparisons.

Table 3
Distribution of functional and declared units by industry sector within the reviewed literature (*N* = 96).

Reference Unit Type	Relative Frequency (%)	Prevalent Sectoral Applications	Representative Reference Cases
Mass-based	> 50%	Agri-food, Manufacturing, Heavy Industry	Zingale et al. (2022, 2024); Guest et al. (2017); Eltohamy et al. (2024)
Volume / Energy-based	21%	Bioenergy, Utilities, Liquid Logistics	Spath and Mann (2000); Koroneos et al. (2008)
Other / Custom	18%	Services, Packaging, Circular Economy	Peñacoba-Antona et al. (2021); Cortesi et al. (2024)

- **Functional Unit (FU):** Quantifies the qualitative and quantitative performance of a product system over a clearly defined temporal lifespan (e.g., “the containment of 1,000 litres of pasteurised milk under refrigerated distribution for 14 days”) (Ross et al., 2017; Cruz-Rivero et al., 2025).
- **Declared Unit (DU):** Represents a simple reference quantity used when the final function of the product or intermediate material is unknown or decoupled from downstream use phases (e.g., “1 kg of unrefined palm oil at the mill gate”).

The data indicates that over half of the manufacturing and agri-food studies rely on mass-based units as standardised reference points. For example, in dairy systems, the mass of milk (e.g., 1 kg of fat- and protein-corrected milk) remains the dominant unit to guarantee alignment with Product Category Rules (PCR). However, from an operational Supply Chain Management (SCM) perspective, treating a simple declared unit (1 kg of material) as a functional unit masks downstream efficiency variation, material degradation, and logistics realities.

4.2.3. *System boundaries and inter-organisational scope lock-ins*

The utility of a functional or declared unit is directly bounded by the structural scope of the life cycle assessment. The reviewed corpus shows a clear division in system boundary selection, split into three distinct operational scopes:

- **Cradle-to-Gate (55%):** Driven by primary data availability and clear corporate boundaries, the majority of SCM studies terminate at the factory gate. This represents a significant structural limitation for Sustainable Supply Chain Management (SSCM). By focusing primarily on production, these studies decouple upstream procurement from downstream distribution, consumer use, and disposal dynamics.
- **Cradle-to-Grave (31%):** Essential for capturing downstream impacts, material recovery potentials, and circular interactions.
- **Gate-to-Gate (14%):** Restricted to isolated manufacturing steps, providing micro-level process visibility but failing to capture broader supply chain dynamics.

This structural configuration creates what can be conceptualized as an operational ‘transparency block’ within multi-tier value chains. When multinational enterprises attempt to evaluate supplier-specific environmental footprints, the widespread combination of mass-based declared units and cradle-to-gate boundaries forces an over-reliance on secondary background datasets. Because databases like Ecoinvent (which represents 78% of the surveyed dataset usage) inherently employ broad regional allocation rules and geographic averages, they routinely mask the unique environmental efficiencies of specific local procurement contexts. Consequently, localized improvements in energy mixes, transport routing, or waste-minimisation protocols implemented by individual suppliers are entirely obscured. To build resilient, auditable,

and genuinely sustainable value chains, future research must shift from arbitrary declared units toward performance-justified functional units that explicitly reflect the dynamic operational boundaries of the entire supply network.

4.3. *The rigour crisis: sensitivity and uncertainty analysis*

The analytical robustness and predictive reliability of life cycle assessment models within Supply Chain Management (SCM) are substantially compromised by a widespread lack of statistical validation. While supply chains are, by their very nature, dynamic, global, and subject to constant operational fluctuations—such as shifting supplier locations, volatile logistical routing, and variations in regional grid electricity mixes—the corresponding literature frequently fails to account for these parameters. A critical methodological gap appears within the reviewed corpus, as almost half (47%) of the 96 analysed studies completely omit both sensitivity analysis (SA) and uncertainty analysis (UA). This systemic omission introduces significant vulnerabilities regarding the credibility of environmental claims and undermines stakeholder confidence in the resulting decision-making frameworks.

Among the limited subset of literature that incorporates these supplementary evaluations, methodological practices vary considerably. Studies that address variability employ sensitivity or uncertainty analysis in roughly equal proportions, accounting for approximately 25% each, whereas a highly restricted portion of the corpus (around 10%) integrates both analytical approaches simultaneously. Furthermore, this lack of methodological consistency exhibits strong sector-specific variations. Manufacturing and industrial studies tend to display a significantly higher frequency of uncertainty analysis, a trend driven by the need to model complex material input variabilities and linear process parameters. Conversely, waste management and policy-related LCA studies frequently omit these verification protocols entirely. This presents a severe paradox, particularly for policy-driven applications where the disclosure of confidence intervals and statistical thresholds is crucial for regulatory communication and macro-level strategy formulation.

Consequently, these findings underscore a pressing need for structural harmonisation across the discipline. Addressing this rigour crisis requires the standardisation of mandatory uncertainty and sensitivity analysis protocols, with a particular emphasis on public sector and policy contexts to ensure regulatory accountability. Furthermore, the development of regionally adapted Life Cycle Impact Assessment (LCIA) methods is essential to mitigate current geographical biases and alleviate data constraints in emerging economies. Finally, the active promotion of open-source software platforms must be prioritised to reduce market monopolisation, thereby increasing methodological flexibility, reproducibility, and data transparency across global supply networks.

5. **Sectoral applications and supply chain integration**

The operationalisation of Life Cycle Assessment (LCA) within supply chain management (SCM) is not uniform; rather, it is highly contingent upon the structural characteristics, regulatory pressures, and material complexities of specific industrial sectors. The compiled text corpus of the 96 selected articles reveals clear sectoral concentrations, with empirical efforts predominantly clustered around agri-food systems and heavy manufacturing or industrial processes. While the underlying mathematical principles of life cycle inventorying remain consistent, the strategic deployment of these metrics—ranging from localized midpoint toxicity concerns to broader global warming potentials—varies significantly to match sector-specific governance demands. Consequently, understanding these sectoral variations is essential for identifying how different industries transform environmental data into actionable procurement strategies, multi-tier supplier benchmarks, and risk-mitigation frameworks. The following subsections provide a comparative, granular analysis of these leading sectors and examine how LCA metrics are

structurally integrated into corporate sourcing decisions.

5.1. Agri-food supply chains: the dominant frontier

The systematic review reveals that agri-food systems represent the most mature application of Life Cycle Assessment (LCA) within supply chain frameworks, accounting for 31.2% of the total article pool. This concentration is primarily driven by mounting regulatory scrutiny and consumer awareness surrounding scope 3 emissions, land-use change, and water depletion in global value chains. Methodologically, this sector exhibits a pronounced path dependency: approximately 70% of agri-food studies rely on the ReCiPe impact assessment method processed via SimaPro software utilising Ecoinvent background datasets. This alignment allows researchers to concurrently evaluate midpoint environmental stressors (such as agricultural runoff and greenhouse gas releases) alongside endpoint metrics related to ecosystem degradation and human health damage.

From a supply chain management perspective, this high concentration of empirical data offers a robust foundation for inter-organisational benchmarking. Landmark studies within this cluster demonstrate how multi-tier transparency can expose environmental hotspots that traditional procurement metrics overlook. For instance, integrated LCA-SCM distribution models highlight the trade-offs between packaging options and food spoilage rates. Similarly, comprehensive assessments across dairy, sheep, and olive oil production lines reveal that upstream farming inputs—such as synthetic fertilisers, herd enteric fermentation, and energy-intensive extraction machinery—frequently overshadow downstream logistics impacts.

However, despite this empirical depth, a critical governance gap remains. Agri-food supply chains are profoundly reliant on external, secondary databases, with less than 5% of studies utilising primary-only inventory data. This heavy reliance on generic dataset averages creates a significant barrier when attempting to benchmark individual suppliers. If a purchasing organisation uses regional or national inventory averages to assess its agricultural inputs, it cannot reward local suppliers who implement superior soil management or carbon-sequestration practices. Furthermore, the almost complete exclusion of social indicators within this sector (found in only 2 out of 30 agri-food studies) creates an operational blind spot. A truly sustainable supply chain network cannot evaluate ecological metrics in isolation; it must integrate worker safety, smallholder livelihoods, and local community impacts into its primary multi-criteria decision-making matrices.

5.2. Industrial and manufacturing supply chains: automotive, textile, and beyond

Industrial and manufacturing supply chains represent the second-largest cluster within the reviewed literature, moving progressively away from purely descriptive footprint analyses toward prescriptive decision tools. In these sectors, supply chain networks feature distinct structural tiers, complex material transformations, and global components sourcing. Methodological choices here deviate significantly from the agri-food standard; manufacturing studies demonstrate a 70% preference for the CML midpoint method over endpoint frameworks. This choice is dictated by the precise material requirements of heavy industrial processes, where engineering and operations teams require granular, scientifically robust visibility into resource depletion, abiotic scarcity, and localized eco-toxicity pathways—such as those associated with metal extraction, chemical synthesis, or electronics fabrication.

A core operational feature of industrial and manufacturing studies is the frequent incorporation of Uncertainty Analysis (UA) and Sensitivity Analysis (SA). Approximately 25% of the reviewed studies within this cluster deploy these stochastic methods to handle the high variability inherent in input materials, fluctuating energy grids, and unpredictable component lifespans. Despite this statistical rigour, manufacturing LCAs are frequently constrained by their tight system boundaries: 55% of the

surveyed literature adheres strictly to a cradle-to-gate scope. For multi-tier supply chains, stopping at the factory gate severely limits long-term strategic value. In complex sectors like automotive assembly or textile production, the environmental burdens associated with specialized surface treatments, global transport legs, consumer use phases, and end-of-life processing are routinely externalized, masking the true-life cycle impact of the network.

To bridge these system boundary gaps and mitigate data deficiencies, advanced manufacturing studies have begun transitioning toward hybrid economic input-output LCA methods. These hybrid frameworks allow purchasing organisations to combine process-level inventory data with macroeconomic trade flows, successfully mapping and quantifying emissions within upstream supply chain tiers that are traditionally obscured by corporate confidentiality or opaque multi-layered global sourcing channels.

5.3. LCA as a strategic tool for sustainable sourcing and procurement

The true integration of LCA into supply chain management occurs when life cycle environmental metrics shift from retrospective sustainability reporting into real-time procurement and supplier selection criteria. The systematic review shows that forward-thinking organisations are increasingly embedding LCA outputs directly into multi-objective optimisation models. In these advanced applications, traditional purchase considerations—such as unit cost, lead time, and quality thresholds—are mathematically weighted against life cycle carbon footprints, water consumption, and resource depletion metrics.

This operational integration fundamentally changes the dynamic of vendor management. Rather than evaluating a tier-1 supplier solely on the environmental impacts of their final assembly plant, procurement teams use cradle-to-gate LCA evidence to scrutinize the supplier's entire upstream network. This comprehensive visibility prevents “burden shifting”—a common supply chain failure where a focal company reduces its direct operational footprint by outsourcing energy-intensive or highly polluting manufacturing steps to less regulated regions or tiers.

However, translating raw LCA datasets into actionable procurement policies presents serious corporate governance challenges. The reviewed literature reveals that 14% of published studies fail to disclose the underlying LCA software tool, and the remaining corpus frequently omits specific parameters such as software versions or precise characterisation models. For a procurement process to be legally defensible, auditable, and fair to all tendering parties, it requires absolute methodological transparency. If the underlying software configurations, background database versions (such as Ecoinvent vs. GaBi), or characterisation models are hidden, the resulting environmental criteria cannot be independently verified. Organisations looking to operationalize LCA in their sourcing strategies must mandate standardised environmental product declarations (EPDs) based on uniform Product Category Rules (PCRs) to ensure baseline equity across the entire supplier network. To support corporate decision-makers in navigating these methodological complexities, Table 4 provides a structured decision guide for tool and method selection tailored to specific supply chain contexts.

6. Emerging paradigms, gaps and future research directions

The systematic analysis of the 96 reviewed articles not only highlights the current boundaries of Life Cycle Assessment (LCA) within supply chain management (SCM) but also exposes critical methodological omissions and structural vulnerabilities that hinder its long-term viability. To transform LCA from a retrospective, single-attribute diagnostic tool into a proactive framework for sustainable supply chain governance, the literature must evolve along three distinct axes. This section synthesises the core empirical gaps identified across the surveyed corpus, outlining a future research agenda that demands the expansion of sustainability indicators, the temporal and architectural evolution of inventory modelling, and the strategic integration of digital

Table 4Distribution of functional and declared units by industry sector within the reviewed literature ($N = 96$).

Supply Chain Context	Recommended Software	Preferred Inventory Dataset	Key Methodological Requirement	Strategic SCM Objective
Agri-Food & Biomass	SimaPro	Ecoinvent / Agribalyse	Multi-functional allocation; land-use change tracking	Scope 3 mitigation; prevention of agricultural burden shifting
Heavy Manufacturing & Automotive	GaBi / SimaPro	Ecoinvent / Sector-specific	Process-based modelling; mandatory uncertainty analysis (UA)	Hotspot identification in multi-tier global component sourcing
Circular Loops & Waste Management	openLCA / SimaPro	Ecoinvent / Primary data	Expanded cradle-to-grave system boundaries; end-of-life recovery metrics	Maximisation of material recirculation and secondary resource credits
Emerging Economies & Low-Resource Sourcing	openLCA	Localized hybrid data (if available) / Ecoinvent	Regionalised characterisation factors; transparency disclosures (14% gap minimisation)	Baseline equity and auditable compliance in vendor selection

Industry 4.0 infrastructures.

6.1. Beyond environmental metrics: social LCA and circular economy integration

A profound limitation identified across the reviewed literature is the severe fragmentation of sustainability metrics, characterised by an over-reliance on environmental indicators to the detriment of social and circular dynamics. Only 21% of the current literature incorporates Social Life Cycle Assessment (S-LCA) principles. This omission means that critical socio-economic factors—including fair wages, child labour safeguards, human rights protections, and gender equity across multi-tiered supplier networks—are systematically excluded from mainstream supply chain assessments (Rafiaani et al., 2018). To rectify this, future research must move away from isolated ecological tracking and actively adopt the [United Nations Environment Programme, 2020S-LCA](#) guidelines. This integration is vital for labour-intensive tiers within agriculture, mining, and textile manufacturing, ensuring that corporate decarbonisation initiatives do not inadvertently exacerbate social inequities or exploit vulnerable workforces.

Similarly, despite the widespread corporate transition toward the Circular Economy (CE), only 31% of the analysed studies integrate circular dimensions such as material recirculation, remanufacturing, or multi-lifecycle reuse loops. The vast majority of supply chain LCAs retain a fundamentally linear modelling logic, calculating impacts on a single-use basis and treating end-of-life phases as final disposal liabilities rather than secondary resource opportunities. This methodological gap can be resolved by standardising the integration of circularity metrics into established LCIA engines. Researchers and practitioners should universally adopt specialized evaluation tools—such as the Plastic Leak Project (PLP) framework for microplastic quantification (Loubet et al., 2022) and the European Commission's Environmental Footprint (EF) method (Foschi et al., 2020)—to ensure that the net environmental savings of circular supply loops are accurately captured, validated, and scaled.

6.2. Methodological evolution: dynamic LCA and hybrid modelling

The prevailing reliance on static life cycle inventories represents a major vulnerability in current sustainable SCM literature. Static LCA configurations capture a frozen snapshot of industrial and logistics systems, completely ignoring temporal fluctuations, seasonal resource shifts, and evolving energy grid mixes. This limitation is particularly disruptive in fast-evolving sectors like bioenergy, biofuels, and renewable energy grids, where background emissions profiles change continuously. Only 21% of the reviewed studies deploy dynamic LCA models capable of incorporating time-sensitive parameters and forward-looking scenario paths. Developing and institutionalising dynamic LCA frameworks is crucial for supply chain managers who must evaluate long-term capital investments or multi-year supplier contracts against shifting decarbonisation trajectories.

Furthermore, the narrow scope of system boundaries remains a

structural barrier to comprehensive accountability, with 55% of studies restricted to cradle-to-gate evaluations. To overcome this truncation, future research must advance hybrid modelling techniques that combine process-based physical inventories with economic input-output (EIO) data. Currently utilised in approximately 20% of studies, hybrid modelling effectively bridges system boundary gaps by utilising macro-economic data to estimate impacts for minor or hidden supply tiers where primary physical data is unavailable. Forcing a shift toward hybrid, cradle-to-grave, and cradle-to-cradle system architectures ensures that downstream distribution, consumer usage, and post-consumer recycling obligations are fully internalised into the firm's strategic planning.

To illustrate these methodological shifts, [Table 5](#) contrasts the current limitations of static approaches against the requirements of the emerging dynamic and hybrid paradigm.

6.3. Industry 4.0 and digitalisation: AI, blockchain, and procurement disclosures

The slow adoption of modern digital tools represents a missed opportunity for the field; the integration of Industry 4.0 enablers into supply chain LCA remains largely speculative across the reviewed literature. The manual collection of lifecycle inventory data is notoriously slow, labour-intensive, and prone to retrospective reporting errors, making it poorly suited for the rapid pace of modern global logistics. To match the speed of commercial operations, future research must focus on coupling LCA mechanisms with artificial intelligence (AI), machine learning algorithms, and real-time Internet of Things (IoT) sensor arrays. AI-driven LCA architectures can automate the data ingestion process, continuously scraping production telemetry, material weights, and transport energy logs to generate automated, real-time life cycle inventories (Shafiq et al., 2024).

To complement automated data collection, blockchain technology presents a robust solution to the data sharing challenges that plague multi-tier supply networks. By deploying cryptographic, distributed

Table 5

Methodological comparison of static versus dynamic and hybrid LCA frameworks.

Methodological Attribute	Current Static Approach	Future Dynamic & Hybrid Paradigm
Temporal Resolution	Fixed emission factors; historic annual averages.	Real-time, time-sensitive emission factors matching grid variations.
System Boundaries	55% Cradle-to-Gate; often truncates complex downstream phases.	Expanded Cradle-to-Grave/ Cradle-to-Cradle; includes post-consumer loops.
Inventory Source	Secondary database dominance; generic regional averages.	Hybrid: IoT sensor telemetry matched with economic input-output modelling.
Uncertainty Handling	47% omit sensitivity/ uncertainty analysis entirely.	Mandated stochastic modelling (e.g., Monte Carlo) across all supply tiers.

ledgers, an organisation can securely track and verify carbon data across multiple tiers of anonymous or highly competitive suppliers. Each participant updates their specific manufacturing or logistics footprint onto an immutable blockchain ledger, maintaining commercial confidentiality while passing verified cumulative environmental data down the line. This infrastructure provides an unalterable audit trail that directly supports statutory environmental product declarations, scope 3 carbon reporting, and corporate sustainability disclosures. Fig. 6 outlines the conceptual flow of this digitally integrated LCA infrastructure.

7. Conclusion

This final section consolidates the overarching insights generated by this systematic literature review, synthesis, and technical benchmarking of 96 peer-reviewed studies. By transitioning from granular analysis to holistic synthesis, the following subchapters provide a comprehensive overview of the empirical findings, scrutinising the current state of tools, methodological dependencies, and structural rigour within the discipline. Furthermore, it outlines the deep theoretical, practical, and policy implications of these insights for industrial networks, concluding with a strategic future research agenda designed to guide the operational evolution of life cycle assessment frameworks toward more dynamic, transparent, and socially responsible global supply chains.

7.1. Summary of empirical findings

This study has provided a systematic and methodologically rigorous overview of the operational integration of Life Cycle Assessment (LCA) frameworks within Supply Chain Management (SCM), auditing an empirical corpus of 96 peer-reviewed articles published up to March 2025. The consolidated evidence reveals that while life cycle metrics are increasingly vital for sustainable supply chain governance, their deployment remains highly uneven. The agri-food sector represents the dominant frontier of this literature, accounting for nearly a third of the reviewed studies, followed closely by manufacturing and circular economy activities such as waste management and bioenergy. Conversely, carbon-intensive and critical infrastructural sectors, including logistics, transportation, energy, and utilities, remain

substantially underexplored within combined LCA-SCM frameworks, marking a significant misalignment between sectoral carbon footprints and academic attention.

Methodologically, the field is governed by powerful institutional and regional path dependencies. SimaPro stands as the primary software platform, and Ecoinvent serves as the dominant secondary data source, characterising more than three-quarters of the surveyed literature. This concentration establishes strict technical couplings, such as the prominent alignment between the ReCiPe method and European software configurations, which severely constrains methodological flexibility. Furthermore, this review exposes a widespread rigour and transparency crisis across global value chains: over half of the empirical studies rely on a truncated cradle-to-gate system boundary, functional units are frequently conflated with simple mass-based declared units, and nearly half of the entire corpus completely omits both sensitivity and uncertainty analyses. This lack of statistical validation and the notable percentage of software version non-disclosure severely undermine the auditable reliability of environmental claims, while emerging frameworks like Social LCA, dynamic inventory modelling, circular loops, and Industry 4.0 digital innovations remain largely speculative and nascent.

7.2. Theoretical, practical, and policy implications

The structural insights generated by this cross-sectoral review establish profound implications for researchers, industrial practitioners, and regulatory policymakers. To translate these overarching lessons into targeted, actionable guidance, Table 6 maps specific recommendations for each stakeholder group based on the predominant supply chain contexts identified within the literature.

By referencing this structured decision matrix, the academic community can deliver a consolidated mapping of the methodological landscape, offering a baseline context that allows future inquiries to explicitly benchmark their inventory choices and justify departures from established technical norms. For corporate practitioners, this cross-sectoral evidence confirms that life cycle frameworks can effectively prevent multi-tier burden shifting; however, the observed lack of data disclosure underscores that managers must enforce the standardised parameters outlined above to ensure fair and verifiable vendor relationships. Finally, for policymakers, this guide provides the robust empirical evidence necessary for designing harmonised regulatory mandates, whilst supporting the urgent case for public investment in localized life cycle datasets to bridge the existing geographical divides.

7.3. Future research priorities

To overcome the structural vulnerabilities exposed in the current literature, the future research agenda must evolve along coordinated methodological and technological axes. Firstly, researchers and practitioners must standardise the technical distinction between functional and declared units under ISO 14040/14044 protocols to ensure consistent cross-study applicability. Addressing the discipline's rigour crisis requires the mandatory enforcement of stochastic uncertainty analysis, such as Monte Carlo simulations, as a baseline reporting requirement for comparative and policy-relevant assessments. Structurally, system boundaries must be forced past the factory gate toward expanded cradle-to-grave and cradle-to-cradle architectures that fully internalise post-consumer recycling obligations and multi-lifecycle loop dynamics.

Furthermore, the narrow ecological focus of traditional supply chain assessments must be balanced by expanding the scope of sustainability indicators. This requires the systematic integration of Social LCA principles, utilising the UNEP guidelines as a minimum benchmark to safeguard human rights and community-level equity across multi-tiered supplier networks. Methodologically, the field must transcend static inventories by developing and validating dynamic LCA frameworks capable of tracking real-time temporal variations in emissions and shifting grid electricity mixes. Ultimately, the scalability and speed of

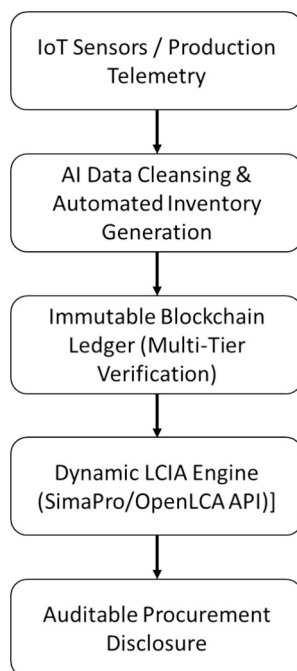


Fig. 6. Conceptual framework for an Industry 4.0-driven supply chain LCA architecture.

Table 6

Decision guide for LCA integration by stakeholder group and supply chain context.

Supply Chain Context	Implications for Researchers (Reviewing Theory)	Implications for Practitioners (Applying Practice)	Implications for Policymakers (Drafting Policy)
Agri-Food & Biomass Networks	Focus on standardising allocation rules for multi-functional systems and tracking regional land-use dynamics.	Mandate primary farm-level data over generic averages to reward suppliers using carbon-sequestration practices.	Formulate Scope 3 emission disclosure rules that account for agricultural seasonality and local farming realities.
Industrial & Manufacturing Sourcing	Advance hybrid modelling (EIO-LCA) to pierce corporate confidentiality and map opaque, multi-tier global supply chains.	Integrate cradle-to-grave LCA metrics into real-time procurement models alongside unit cost and lead time.	Establish harmonised standards for Environmental Product Declarations (EPDs) to prevent cross-border burden shifting.
Circular Economy & Waste Systems	Standardise the integration of circularity engines into established LCIA tools to capture material recovery loops accurately.	Shift from a cradle-to-gate scope to expanded post-consumer boundaries to capture long-term material recirculation credits.	Design regulatory frameworks and economic incentives that mandate extended producer responsibility (EPR) metrics.
Emerging Economies (Global South)	Develop regionalised characterisation factors to reduce the systemic methodological reliance on European baselines.	Adopt open-source platforms (e.g., openLCA) whilst actively auditing the 14% software version non-disclosure gap for transparency.	Target public funding toward the creation of domestic life cycle inventory (LCI) databases matching local energy mixes.

these frameworks depend on the strategic adoption of Industry 4.0 digital enablers. Future research should actively investigate the deployment of automated, artificial intelligence-driven data ingestion mechanisms and cryptographic blockchain infrastructures to construct immutable, transparent, and legally defensible environmental disclosure channels across global supply chains..

CRedit authorship contribution statement

Busola Akintayo: Writing – original draft, Visualization, Investigation, Formal analysis, Data curation, Conceptualization. **Angel M. Gento:** Writing – review & editing, Validation, Supervision, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Damilola Caleb Akintayo:** Visualization, Supervision, Software, Methodology, Investigation. **Yesica Pino:** Writing – review & editing, Visualization, Supervision, Project administration, Methodology, Investigation. **Oludolapo Olanrewaju:** Writing – review & editing, Visualization, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.clscn.2026.100371>.

Data availability

The datasets generated and analysed has been included in the main manuscript. The authors can make further material through reasonable requests.

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